Taewoong Yoon

Department of Physics and Astronomy, Seoul National University
1 Gwanak-ro, Gwanak-gu, Seoul 08826, Korea
twyoon@snu.ac.kr · twyoon.github.io

EDUCATION

Seoul National University (SNU) Mar 2019 – Feb 2027 (expected)
Ph.D. in Physics · Advisor: Prof. Hyunyong Choi · Seoul, Korea

Pohang University of Science and Technology (POSTECH) Mar 2014 – Feb 2019
B.S. in Physics · Pohang, Korea

RESEARCH EXPERIENCE

Research Assistant, Korea Institute of Science and Technology (KIST) Feb 2026 – Present
Advisor: Dr. Junghyun Lee

- Estimated defect densities from DEER measurements in a collaboration between KIST and the group of Prof. David Awschalom at the University of Chicago.
- Migrated the control software of an NV measurement setup from LabVIEW to Python, including FPGA-based timing and AWG pulse generation.
- Performing Ramsey measurements on a single NV center toward a proof-of-principle demonstration of postselected metrology using the NV spin qutrit.

Graduate Research Assistant, Seoul National University (SNU) Aug 2019 – Present
Advisor: Prof. Hyunyong Choi

- Polarized the optically dark P1 electron spin bath to 740 times its thermal-equilibrium value via iterative Hartmann–Hahn transfer from NV centers, and characterized its collective coherence and disorder.
- Demonstrated controlled polarization transfer between spectrally resolved P1 subensembles by extending the NV–P1 protocols to interactions within the P1 ensemble.
- Enabled vector magnetometry without a calibrated bias field by identifying NV crystallographic axes from spatially varying microwave fields.
- Created site-controlled single color centers in diamond and hBN, confirmed by photon antibunching, using a home-built femtosecond laser-writing system with SLM-based aberration correction.
- Built the group's first NV measurement system from scratch, including the confocal microscope, microwave delivery, pulse sequencing, and Python-based instrument control and data acquisition, which served as the platform for all experiments above.

PUBLICATIONS

[*] *Equal contribution.* [†] *Corresponding author.*

1. **T. Yoon**, D. Kim, J. Lee[†], and H. Choi[†], "Controlled polarization exchange between dark electron spin subensembles in diamond." *In preparation.*
2. M. M. McMillan[*], H. Kim[*], J. C. Marcks, **T. Yoon**, B. Pingault, N. Delegan, F. J. Heremans, H. Seo[†], J. Lee[†], and D. D. Awschalom[†], "Quantum decoherence in a mixed-dimensional spin bath." *In preparation.*
3. **T. Yoon**, S. Oh, J. Lee[†], and H. Choi[†], "Mesoscopic spin coherence in a disordered dark electron spin ensemble," arXiv:2602.17074 (2026). *Under review, Physical Review Applied.*
4. **T. Yoon**, M. Cha, D. Kim, and H. Choi[†], "Identifying NV center axes via spatially varying microwave fields for vector magnetometry," *Applied Physics Letters* **126**, 144002 (2025).

PRESENTATIONS

1. **APS Global Physics Summit**, Denver, USA, Mar 2026. *(oral)*
   "Polarization of P1 electron spin bath via repetitive Hartmann-Hahn transfer"
2. **KPS Spring Meeting**, Daejeon, Korea, Apr 2024. *(oral)*
   "Probing and controlling many-body dipolar interactions between electron spins in diamond"
3. **Optics and Photonics Congress (OPC)**, Jeju, Korea, Aug 2023. *(oral)*
   "Precise calibration of magnetic field vector in a diamond nitrogen-vacancy center ensemble"

4. **Conference on Lasers and Electro-Optics (CLEO)**, San Jose, USA, May 2022. *(poster)*
   "Deterministic creation of single nitrogen-vacancy center in diamond using femtosecond laser writing"
5. **Optics and Photonics Congress (OPC)**, Jeju, Korea, Jul 2021. *(oral)*
   "Aberration correction of femtosecond lasers for deterministic quantum emitters in nitrogen-vacancy diamonds"

TECHNICAL SKILLS

Coherent spin control — Pulse-sequence design on AWGs, TTL pulse generators, and FPGAs, including IQ-modulated microwave pulse shaping and synchronization of laser and photon-counting gates.

Optical measurement and control — Scanning confocal microscopy, gated and time-correlated photon counting, photoluminescence spectroscopy, AOM double-pass laser pulse control, femtosecond laser writing with SLM-based aberration correction.

Instrument control and data acquisition — Measurement automation in Python and LabVIEW.

Simulation and modeling — Spin dynamics and Monte Carlo simulations in Python (QuTiP), electromagnetic field modeling (COMSOL).

Fabrication — Photolithography, RIE, thermal evaporation, 2D material exfoliation and transfer.

Design — PCBs, photomasks, and machined mechanical parts (Altium, KLayout, AutoCAD, Fusion 360).

TEACHING EXPERIENCE

Teaching Assistant, Seoul National University
Electrodynamics I (Spring 2021) · Computational Physics (Fall 2020) · Electromagnetic Waves and Optics (Spring 2020) · Physics Lab. 2 (Summer 2019, Fall 2019) · Physics Lab. 1 (Spring 2019)

HONORS AND AWARDS

- BK Excellent Teaching Assistant Award — Jul 2020
- BK Frontier Fellowship — Mar 2019 – Aug 2019
- National Scholarship for Science and Engineering — Mar 2014 – Feb 2018